SAKSHAM GUPTA

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EDUCATION

Manipal Institute of Technology, Manipal — ***Manipal, Karnataka • May 2027***

B.Tech** in **Information Technology

CGPA: 9.16/10.0

Delhi Public School R.K. Puram — ***New Delhi • 2023***

Class 12 – CBSE (90.2%)

Class 10 – CBSE (93.5%)

EXPERIENCE

Software Development Intern — ***June – July, 2025***

Indian Railway Catering and Tourism Corporation (IRCTC)

- Analyzed enterprise booking system architecture handling 2M+ daily transactions, identifying data flow and performance bottlenecks
- Worked closely with relational databases to analyze query execution paths and optimize database query performance by 25%
- Documented database schemas, booking workflows, and data dependencies for system optimization initiatives
- Evaluated scalable system design patterns including microservices architecture for high-traffic applications
- Assisted in automating database performance and load tests using Python scripts to simulate concurrent transactions

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, C, PHP

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, EDA & Visualization (Matplotlib, Seaborn), Data Analysis, Model Evaluation & Validation

Databases, Tools & Security: SQL, MongoDB, Git, VS Code, IntelliJ IDEA

Web Technologies: Flask, Node.js, React, REST APIs, HTML/CSS

Systems & Embedded: ARM Cortex-M3, Interrupt Handling, Linux, Shell Scripting, Windows

Software Engineering: SDLC, Agile Methodology, Requirements Engineering, Software Testing, UML Diagrams, Risk Management

PROJECTS

SafePay | Machine Learning Based Transaction Fraud Detection System | GitHub • View Live

- Built an ML-based fraud detection model with data preprocessing and feature engineering, achieving 85%+ fraud detection accuracy
- Integrated real-time model inference into a Python backend (Flask) with MongoDB for transaction validation
- Implemented end-to-end security using AES-256 encryption, digital signatures, and secure password hashing

Starfinder | Multi-Class Stellar Classification System | GitHub • View Live

- Developed multi-class stellar classification using Random Forest, achieving 93% accuracy across 6 star types
- Implemented domain-aware validation pipeline to detect out-of-distribution inputs and prevent invalid compact object predictions
- Deployed production ML system with Flask REST API, automated model persistence, and comprehensive EDA pipeline

Code2Flow | AI Code-to-Flowchart Generator | GitHub • View Live

- Created an AI-powered tool that converts source code into deterministic ASCII-based flowcharts using Python (Flask) and the Gemini API
- Designed a hybrid pipeline separating LLM-based control-flow extraction from deterministic Python rendering for stable output
- Established keyword-based normalization and validation to handle unreliable LLM formatting and minimize API usage

Fileshare | Real-Time File & Text Sharing | GitHub • View Live

- Engineered a real-time room-based file and text sharing app using Node.js, Express, and WebSockets
- Created ephemeral rooms and auto-expiring messages/files for privacy and low storage usage
- Supported file uploads (≤10MB) with Multer and optional PostgreSQL-backed room metadata

LEADERSHIP & ACTIVITIES

Organizing Committee Member — Sports (Chess) — ***March 2025***

Revels'25, MIT Manipal

- Led logistics coordination for chess tournaments with 200+ participants, managing registration and bracket systems
- Collaborated with a 12-member cross-functional team ensuring 100% on-time event execution during cultural fest

ADDITIONAL

Languages: Hindi (Native), English (Proficient)

Interests: Competitive Programming, Chess, Badminton, Reading

ACHIEVEMENTS

Achievers Scholarship (Top 5% in IT)

Active contributor to open-source projects